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STUDY ON MAIN REPRODUCTIVE STEROIDAL HORMONES IN POSTMENOPAUSAL BREAST CANCER PATIENTS

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ABSTRACT

In the present work, thirty nine (39) patients were randomly selected from Radiotherapy Department of Jinnah Post Graduate Medical Center, Karachi. The criteria of inclusion was, females of postmenopausal age, biopsy proven breast cancer (unilateral or bilateral), mastectomy with auxiliary clearance within fifteen days, without radiation or hormonal therapy and no clinical metastasis was present and twelve postmenopausal controls were also selected, for comparison. This study leads to the discovery that there is a co-relation of estrogen and progesterone in breast cancer women in Karachi, and this relationship is reciprocal, estrogen level increases with decrease in estrogen receptor showing aggressiveness in disease.

INTRODUCTION

The use of oral contraceptives may induce breast cancer (Stoll, 1964). Women with breast cancer excreted less estradiol (E_3) as compared to estrone (E_1) and estradiol (E_2) (Cole et al., 1976). Elevated urinary (E_3) in women with breast cancer (high risk group) and higher levels of both (E_2) and prolactin are notable at comparable times of the menstrual cycle than control population (Hellman et al., 1971 and Zumoff et al., 1975). Estrogen regulates protein and cell growth in human breast cancer (Cullen et al., 1989). Remission of mammary carcinomas under hormone therapy requires the tumors to be positive for estrogen receptors, whereas a higher degree of remission occurred if the tumors were additionally positive for androgen and progesterone receptors (Caffier, 1982). Correlation of estrogen receptor status of mammary carcinoma metastasis with the free interval and survival time in sequential endocrine therapy and chemotherapy is cited in the literature (Stolzenbach et al., 1982). The estrogenic stimulation of breast cell growth in the absence of sufficient cyclic progesterone secretion may provide a setting favourable for the development of breast cancer (Sherman and Koreman, 1974). The hormone and cytotoxic drug responsiveness of cultured human breast cancer cells are resistant to specific hormones is also investigated (Coldham et al., 1990).

The clinical pharmacology and use of progestins and progesterone in the treatment of postmenopausal symptoms (Whitechead et al., 1983). The mechanism involved in the evaluation of progestin resistance in human breast cancer cells has been investigated (Murphy et al., 1991). The cytochemistry of sex steroid receptors specially estrogen receptors in breast cancer is also being reviewed (McCarty et al., 1981).

MATERIAL AND METHODS

Patients and Controls:

In this study thirty nine (39) blood samples were collected from postmenopausal women suffering from breast cancer while twelve (12) blood samples collected from postmenopausal women who were not suffering from any disease. Both group of women were residents of Karachi.

Laboratory materials:

All tests were performed by radioaimmunoassay procedure and all components kept at room temperature before use. Materials used for performing the test were serum estradiol/progesterone, Ab-coated tubes, one vial ^{125}I-serum estradiol/ progesterone liquid (105) ml, serum estradiol/progesterone calibrators, Vortex mixer (USA), Gamma Counter (UK), plain poly propylene tubes, micropipettes, incubation bath, foam decanting rack and tri-Level human serum based immuno-assay control.

ASSAY OF SERUM ESTRADIOL

Method:

Assay carried out was comprised of no-extraction solid phase ^{125}I-radioimmunoassay quantitative coat-A-count serum estradiol measurement technique. The coat-A-count serum estradiol procedure was based on antibody-coated tubes ^{125}I-labelled serum estradiol which competes with serum estradiol in the patient sample for antibody sites. After 1-3 hours incubation at 37°C on a shaker, separation of bound form free was achieved by simply decanting the material. The tube was then counted in a gamma-counter, the counts being inversely related to the amount of serum estradiol present in the patient sample. The quantity of serum estradiol in the sample was determined by comparing the counts to a standard curve (human serum-based) standards having serum estradiol values ranging from 20 to 3600 pg/ml ≈ 0.07-13.2 (n/mol/lit). Total counts approximately of 50,000 cpm at iodination were determined as corresponding to a maximum binding of 35-45%. The sensitivity of the assay was temperature dependent.

ASSAY OF SERUM PROGESTERONE

Method:

Assay method employed was a no-extraction, solid phase ^{125}I-radioimmunoassay designed for the measurement technique. The antibody being immobilized to the wall of the polypropylene tube, decanting the supernatant suffices to terminate the competition and isolated the antibody-bound fraction of the radiolabelled serum progesterone, at 37°C. Counting the tube in a gamma counter yields a number, which converts by way of calibration of the serum progesterone present in the patient sample. The hundred (100) tube serum progesterone kit equipped with human serum based standards gave serum progesterone values ranging from 0.1 to 40 mg/ml (0.3-127 n/mol/lit.). The calibrators were supplied in liquid form. The tracer has a specific activity with total counts of approximately 75,000 cpm at iodination. Maximum binding was found to be approximately 40%.

RESULTS AND DISCUSSION

Thirty nine (39) patients were randomly selected from Jinnah Post Graduate Medical Center, Karachi, while twelve (12) human subjects of the same age, having no medical or psychiatric illness and not using any hormonal preparation were taken as control. Serum estradiol and progesterone status of both the patients and control group were determined by radioimmunoassay technique.

The statistical data on postmenopausal serum estradiol and serum progesterone levels in control subjects and in cancer patients are represented in Table 1. The standard normal value of serum estradiol in postmenopausal women reported by Diagnostic Product Corporation (DPC) USA is 14 pg/ml and normal value of serum progesterone is 0.1 ng/ml.

The Table-1 display two groups, patients (39) and control (12). The level of estradiol is 18.66 ± 6.31 pg/ml in controls and 54.56 ± 10.20 pg/ml in cancer patients. The Table 2 shows the relationship of progesterone (ng/ml) to different levels of estradiol (pg/ml). The Table 3 shows the relationship of estradiol different levels of progesterone. The Table 4 shows the estradiol receptor status (f. mole/mg) versus lymphnode involvement (complete, partial and nil). The Table 5 shows lymphnode status with respect to the metastasis in patients with different levels of estradiol and progesterone.

CONCLUSION

a) This study support that high estradiol and progesterone levels are among the risk factors of breast cancer in Karachi women, as claimed by other investigators.

b) The study shows that normal physiological reciprocal relationship between estradiol and progesterone exist in breast cancer patients.

c) This study show that as estradiol level increases, sensitivity of estrogen receptor decreases, and disease becomes aggressive showing metastatic involvement of a greater number of nodes.

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Table 1
Estradiol and progesterone levels in controls and patients

Group	Estradiol (pg/ml)°	Progesterone (ng/ml)°
Control	18.66 ± 6.31 (12)	0.229 ± 0.01 (12)
Patients	54.56 ± 10.20 (39)⁺*	0.170 ± 0.01 (39)

⁺ Mean + S.E. (No. of Samples)
* $P < 0.05$
° pg = 10^{-12} g, ng-10^{-9} g

Table 2
Relationship of progesterone to different level of Estradiol

Estrogen (pg/ml)	≤ 15	≤ 25	≤ 75	≤ 125	> 125
Progesterone (ng/ml)	0.15 (11)	0.18 (11)	0.16 (09)	0.178 (06)	0.167 (04)

Mean (Total number of patients)

Table 3
Relationship of Estradiol to different levels of Progesterone

Progesterone (pg/ml)	≤ 0.1	≤ 0.15	≤ 0.20	> 0.20
Estradiol (ng/ml)	32 (14)	95.5 (10)	30.16 (06)	59.77 (09)

Mean (Total number of patients)

Table 4
Estradiol receptor status vs lymph nodes involvement

Estradiol receptor status (f mole/mg)	Lymph nodes involvement	
≤ 100	Complete	-02 (40 %)
	Partial	-02 (40 %)
	Nil	-01 (20 %)
> 100	Complete	- 01 (12.5 %)
	Partial	- 03 (37.5 %)
	Nil	- 04 (50.0 %)

Table 5
Lymph Nodes

Lymph node status	Estradiol (pg/ml)	Progesterone (ng/ml)
All nodes show metastasis	32.2 ± 6.8 (7)	0.188 ± 0.04 (7)
Some nodes show metastasis	65.0 ± 17.2 (13)	0.160 ± 0.019 (13)
Nodes without metastatus	60.0 ± 23.0 (13)	0.190 ± 0.029 (13)

Mean (Total number of patients)
$p > 0.05$ in all cases

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